Peaks of the non-Gaussian compaction function for broad spectra

Phase 3b derivation (Option A notes)

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Abstract

Peaks theory with local non-Gaussianity (NG) has so far imposed the peak condition on a Gaussian field (the Gaussian curvature $\zeta_{\rm g}$, $-\Delta\zeta_{\rm g}$, or the Gaussian part of the compaction $C_{\rm g}$) and mapped the result through F. For the compaction function this is exact only when the spectrum is narrow. Here the peak condition is imposed on the non-Gaussian linear compaction $C_\ell = F'(\zeta_r)\, C_{\rm g}$ itself, which is a local function of *two* correlated Gaussian fields. The result is compact: at a point with $(C_{\rm g}, \zeta_r) = (c, z)$, maxima of C_ℓ are maxima of the Gaussian field $Y_\lambda = C_{\rm g} + \lambda\zeta_r$ with $\lambda = cF''(z)/F'(z)$, corrected by a rank-one Hessian term proportional to $\kappa = -cF'(1/F')''$. The peak density is the BBKS expression for Y_λ, which reduces to the Gaussian-peak approximation (Phase 3a) when $\lambda = 0$ or the spectrum is monochromatic, times a factor $1 - \tau R_2$ for the rank-one term that is exact to first order. That factor is identically 1 for the logarithmic (ultra-slow-roll) template, below 4% for $|f_{\rm NL}| \le 2$ at $\Delta \le 0.3$, and up to tens of percent at individual nodes for $f_{\rm NL} = -2$ at $\Delta = 1$. An independent Monte Carlo of the twenty-variable Gaussian confirms both pieces.

1 Setup

The curvature perturbation is $\zeta(\boldsymbol{x}) = F(\zeta_{\rm g}(\boldsymbol{x}))$ with $\zeta_{\rm g}$ Gaussian, power spectrum $\mathcal{P}(k)$. At smoothing scale r define two Gaussian fields,

$$C_{\rm g}(\boldsymbol{x}; r) = \int \frac{{\rm d}^3k}{(2\pi)^3}\, e^{i\boldsymbol{k}\cdot\boldsymbol{x}} K_c(k)\, \zeta_{\rm g}(\boldsymbol{k}), \qquad K_c(k) = \tfrac{2\Phi}{3}(kr)^2 W(kr)\, T(kr), \tag{1}$$

$$\zeta_r(\boldsymbol{x}; r) = \int \frac{{\rm d}^3k}{(2\pi)^3}\, e^{i\boldsymbol{k}\cdot\boldsymbol{x}} K_r(k)\, \zeta_{\rm g}(\boldsymbol{k}), \qquad K_r(k) = W_s(kr)\, T(kr), \tag{2}$$

with $W(y) = 3(\sin y - y\cos y)/y^3$, $W_s(y) = \sin y/y$, the radiation transfer function $T(y) = W(y/\sqrt{3})$ at $\tau = r$, $\Phi = 2/3$, and optionally a hard cut $K_{c,r} \to 0$ for $kr > y_{\rm cut}$. For a configuration that is spherically symmetric about $\boldsymbol{x}$, $C_{\rm g} = -2\Phi r\zeta_{\rm g}'(r)$ and $\zeta_r = \zeta_{\rm g}(r)$, so the linear compaction of $\zeta = F(\zeta_{\rm g})$ is [1, 2]

$$C_\ell(\boldsymbol{x}; r) = F'\big(\zeta_r(\boldsymbol{x}; r)\big)\, C_{\rm g}(\boldsymbol{x}; r), \qquad C = C_\ell - \frac{C_\ell^2}{4\Phi}. \tag{3}$$

Threshold statistics integrate the bivariate Gaussian $P_{\rm G}(c, z)$ of $(C_{\rm g}, \zeta_r)$ over the type-I domain $\mathcal{D} = \{C > C_{\rm th},\ C_\ell < 2\Phi\}$: $\beta_{\rm th} = \int_{\mathcal{D}} {\rm d}c\, {\rm d}z\, P_{\rm G}(c, z)\, \mathcal{K}(C - C_{\rm th})^\gamma$. Peaks theory (prescription (i) of the gap assessment) counts local maxima in $\boldsymbol{x}$ of the field $C_\ell(\boldsymbol{x}; r)$ at fixed r. Phase 3a counted maxima of $C_{\rm g}$ instead (minima where $F' < 0$); here we count maxima of C_ℓ.

2 What is counted

By the Kac–Rice formula, the number density of maxima of C_ℓ with $(C_\mathrm{g}, \zeta_r) \in \mathrm{d}c\,\mathrm{d}z$ is

$$n(c,z)\,\mathrm{d}c\,\mathrm{d}z = \Big\langle \delta_\mathrm{D}(C_\mathrm{g}-c)\,\delta_\mathrm{D}(\zeta_r - z)\,\delta_\mathrm{D}^3(\nabla C_\ell)\,\big|\det \nabla\nabla C_\ell\big|\,\Theta(-\nabla\nabla C_\ell \succ 0)\Big\rangle\,\mathrm{d}c\,\mathrm{d}z, \tag{4}$$

and the mass fraction is

$$\beta_\mathrm{pk} = V\int_\mathcal{D} \mathrm{d}c\,\mathrm{d}z\; n(c,z)\,\mathcal{K}(C-C_\mathrm{th})^\gamma, \qquad V = \tfrac{4\pi}{3}r^3, \tag{5}$$

with the same domain, threshold, window, transfer function and cut as β_th.

3 Gaussian statistics at a point

For $A, B \in \{C_\mathrm{g}, \zeta_r\}$ define $\sigma^2_{n,AB} = \int \mathrm{d}\ln k\; k^{2n} K_A K_B\,\mathcal{P}$. Homogeneity and isotropy give

$$\begin{aligned} \langle AB\rangle &= \sigma^2_{0,AB}, \qquad \langle \partial_i A\,\partial_j B\rangle = \tfrac{1}{3}\delta_{ij}\sigma^2_{1,AB}, \qquad \langle A\,\partial_i\partial_j B\rangle = -\tfrac{1}{3}\delta_{ij}\sigma^2_{1,AB},\\ \langle \partial_i\partial_j A\,\partial_k\partial_l B\rangle &= \tfrac{1}{15}\sigma^2_{2,AB}(\delta_{ij}\delta_{kl}+\delta_{ik}\delta_{jl}+\delta_{il}\delta_{jk}), \qquad \langle A\,\partial_i B\rangle = \langle \partial_i A\,\partial_j\partial_k B\rangle = 0. \end{aligned} \tag{6}$$

The twenty variables $(C_\mathrm{g}, \zeta_r, \nabla C_\mathrm{g}, \nabla\zeta_r, \nabla\nabla C_\mathrm{g}, \nabla\nabla\zeta_r)$ therefore split into two independent blocks: the six gradient components, and the two scalars with the twelve Hessian components. In the code's notation $\sigma^2_{0,cc} = \sigma^2_c$, $\sigma^2_{0,cr} = \sigma^2_{cr}$, $\sigma^2_{0,rr} = \sigma^2_r$, $r^2\sigma^2_{1,cc} = s^2_1$, $r^4\sigma^2_{2,cc} = s^2_2$, $r^2\sigma^2_{1,cr} = s^2_{cr1}$; Phase 3b needs three more,

$$r^2\sigma^2_{1,rr} = \int \mathrm{d}\ln k\, x^2 W_s^2 T^2\mathcal{P}, \quad r^4\sigma^2_{2,cr} = \tfrac{2\Phi}{3}\int \mathrm{d}\ln k\, x^6 W W_s T^2\mathcal{P}, \quad r^4\sigma^2_{2,rr} = \int \mathrm{d}\ln k\, x^4 W_s^2 T^2\mathcal{P}, \tag{7}$$

with $x = kr$. All three converge with the transfer function; $\sigma^2_{2,cc}$ diverges logarithmically without the cut, so the cut stays (for both statistics).

4 Derivatives of the non-Gaussian compaction

Differentiating (3),

$$\partial_i C_\ell = F'\,\partial_i C_\mathrm{g} + F''\,C_\mathrm{g}\,\partial_i\zeta_r, \tag{8}$$

$$\partial_i\partial_j C_\ell = F'\,\partial_i\partial_j C_\mathrm{g} + F''\,C_\mathrm{g}\,\partial_i\partial_j\zeta_r + F''\big(\partial_i C_\mathrm{g}\,\partial_j\zeta_r + \partial_j C_\mathrm{g}\,\partial_i\zeta_r\big) + F'''\,C_\mathrm{g}\,\partial_i\zeta_r\,\partial_j\zeta_r, \tag{9}$$

with $F^{(n)}$ evaluated at ζ_r. At a point where $(C_\mathrm{g}, \zeta_r) = (c, z)$ define

$$\lambda \equiv \frac{c\,F''(z)}{F'(z)}, \qquad Y_\lambda(\boldsymbol{y}) \equiv C_\mathrm{g}(\boldsymbol{y}) + \lambda\,\zeta_r(\boldsymbol{y}), \tag{10}$$

with λ frozen at its value at the point, so that Y_λ is a Gaussian field. Then $\nabla C_\ell = F'(z)\,\nabla Y_\lambda$: the stationary points of C_ℓ are stationary points of Y_λ, *not* of C_g. There $\nabla C_\mathrm{g} = -\lambda\nabla\zeta_r$ and

$$\nabla\nabla C_\ell = F'(z)\Big[\nabla\nabla Y_\lambda + \kappa\,\nabla\zeta_r\,\nabla\zeta_r^{\,T}\Big], \qquad \kappa \equiv \frac{c}{F'}\Big(F''' - \frac{2F''^2}{F'}\Big) = -c\,F'(z)\,\Big(\frac{1}{F'}\Big)''(z). \tag{11}$$

For the templates in the code:

- quadratic, $F = \zeta_\mathrm{g} + \frac{3}{5}f_\mathrm{NL}\zeta_\mathrm{g}^2$: $\lambda = \frac{6}{5}f_\mathrm{NL}c/(1+\frac{6}{5}f_\mathrm{NL}z)$ and $\kappa = -2\lambda^2/c$;
- logarithmic (USR), $F = -\mu_*\ln(1-\zeta_\mathrm{g}/\mu_*)$: $1/F' = 1 - z/\mu_*$ is linear, so $\kappa \equiv 0$ and $\lambda = c/(\mu_* - z)$;
- curvaton: λ and κ from F', F'', F''' numerically.

5 The gradient constraint

Since $\nabla C_\ell = F'(z)\nabla Y_\lambda$, $\delta_{\rm D}^3(\nabla C_\ell) = |F'(z)|^{-3}\delta_{\rm D}^3(\boldsymbol{u})$ with $\boldsymbol{u} \equiv \nabla Y_\lambda = \nabla C_{\rm g} + \lambda\nabla\zeta_r$. The gradient block is independent of the scalars and Hessians, and (6) gives $\langle u_i u_j\rangle = \frac{1}{3}\delta_{ij}\sigma_{1,\lambda}^2$ with

$$\sigma_{n,\lambda}^2 \equiv \sigma_{n,cc}^2 + 2\lambda\,\sigma_{n,cr}^2 + \lambda^2\sigma_{n,rr}^2, \qquad P(\boldsymbol{u}=0) = \Big(\frac{3}{2\pi\sigma_{1,\lambda}^2}\Big)^{3/2}. \tag{12}$$

Conditional on $\boldsymbol{u}=0$, $\boldsymbol{g} \equiv \nabla\zeta_r$ is Gaussian with zero mean and covariance $v\,\mathbb{I}$,

$$v = \frac{\sigma_{1,cc}^2\sigma_{1,rr}^2 - \sigma_{1,cr}^4}{3\,\sigma_{1,\lambda}^2}, \tag{13}$$

and $\nabla C_{\rm g} = -\lambda\boldsymbol{g}$. The factor $|F'|^3$ from $|\det\nabla\nabla C_\ell|$ cancels $|F'|^{-3}$, and

$$n(c,z) = P_{\rm G}(c,z)\Big(\frac{3}{2\pi\sigma_{1,\lambda}^2}\Big)^{3/2}\mathbb{E}\Big[\big|\det(H_\lambda + \kappa\,\boldsymbol{g}\boldsymbol{g}^T)\big|\,\Theta_s(H_\lambda + \kappa\,\boldsymbol{g}\boldsymbol{g}^T)\,\Big|\,c,z\Big], \tag{14}$$

where $H_\lambda = \nabla\nabla Y_\lambda$, $\boldsymbol{g} \sim \mathcal{N}(0, v\mathbb{I})$ independent of H_λ, $s = \operatorname{sign}F'(z)$, and Θ_+ (Θ_-) requires a negative- (positive-) definite matrix: where $F' < 0$ maxima of C_ℓ are minima of the bracket. Equation (14) is exact within the setup of Sec. 1.

6 Leading order: BBKS peaks of Y_λ

Set $\kappa = 0$. H_λ is the Hessian of the Gaussian field Y_λ; by isotropy the scalars correlate only with its trace,

$$b_c \equiv \langle -\nabla^2 Y_\lambda\, C_{\rm g}\rangle = \sigma_{1,cc}^2 + \lambda\sigma_{1,cr}^2, \qquad b_r \equiv \langle -\nabla^2 Y_\lambda\,\zeta_r\rangle = \sigma_{1,cr}^2 + \lambda\sigma_{1,rr}^2, \tag{15}$$

and the traceless part is independent of them, with the BBKS normalisation set by $\sigma_{2,\lambda}^2$. With $x \equiv -\nabla^2 Y_\lambda/\sigma_{2,\lambda}$, given (c,z), $x \sim \mathcal{N}(m_\lambda, s_\lambda^2)$,

$$m_\lambda = \frac{(b_c,b_r)\,\Sigma^{-1}(c,z)^T}{\sigma_{2,\lambda}}, \qquad s_\lambda^2 = 1 - \frac{(b_c,b_r)\,\Sigma^{-1}(b_c,b_r)^T}{\sigma_{2,\lambda}^2}, \qquad \Sigma = \begin{pmatrix}\sigma_{0,cc}^2 & \sigma_{0,cr}^2\\ \sigma_{0,cr}^2 & \sigma_{0,rr}^2\end{pmatrix}. \tag{16}$$

BBKS (their eqs. A14–A19) give $\mathbb{E}[|\det H|\Theta_+\,|\,x] = \sigma_{2,\lambda}^3 f(x)/27$, with $f(x) \to x^3 - 3x$ for $x \gg 1$. Hence

$$\boxed{n(c,z) = P_{\rm G}(c,z)\,\frac{G(s\,m_\lambda, s_\lambda)}{(2\pi)^{3/2}R_{*,\lambda}^3}, \qquad R_{*,\lambda} = \frac{\sqrt{3}\,\sigma_{1,\lambda}}{\sigma_{2,\lambda}}, \qquad G(m,s) = \int_0^\infty {\rm d}x\, f(x)\,\mathcal{N}(x;m,s^2).} \tag{17}$$

This is the Phase 3a expression with $C_{\rm g} \to Y_\lambda$. In normalised variables $\nu = c/\sigma_{0,c}$, $\hat{z} = z/\sigma_{0,r}$, $m_\lambda = a_\lambda\nu + b_\lambda\hat{z}$ with $a_\lambda = (\gamma_\lambda - \rho\gamma_{z,\lambda})/(1-\rho^2)$, $b_\lambda = (\gamma_{z,\lambda} - \rho\gamma_\lambda)/(1-\rho^2)$, $s_\lambda^2 = 1 - a_\lambda\gamma_\lambda - b_\lambda\gamma_{z,\lambda}$, where $\gamma_\lambda = b_c/(\sigma_{2,\lambda}\sigma_{0,c})$, $\gamma_{z,\lambda} = b_r/(\sigma_{2,\lambda}\sigma_{0,r})$ and $\rho = \sigma_{0,cr}^2/(\sigma_{0,c}\sigma_{0,r})$. At fixed λ these coefficients and $R_{*,\lambda}$ do not depend on the amplitude of $\mathcal{P}$.

Limits:

1. $\lambda = 0$ (Gaussian ζ, or $F'' = 0$): Phase 3a exactly.
2. Monochromatic spectrum: K_c and K_r are evaluated at one k, so $Y_\lambda \propto \zeta_r$ and $C_{\rm g} \propto \zeta_r$; $R_{*,\lambda} = \sqrt{3}/k_*$ and the peak positions do not depend on λ. 3b and 3a coincide. The difference is a finite-width effect.
3. $F'(z) \to 0$ at fixed c: $|\lambda| \to \infty$, $Y_\lambda/\lambda \to \zeta_r$, the peak condition becomes $\nabla\zeta_r = 0$, and $R_{*,\lambda} \to \sqrt{3}\sigma_{1,rr}/\sigma_{2,rr}$, m_λ, s_λ stay finite.

7 The rank-one term

Work in units of $\sigma_{2,\lambda}$ and take $s = +1$ (for $s = -1$ replace $H \to -H$, $\kappa \to -\kappa$, $m_\lambda \to -m_\lambda$). The matrix determinant lemma gives, for $H \prec 0$, $\det(-H - \kappa \boldsymbol{g}\boldsymbol{g}^T) = \det(-H)\,\big(1 - \kappa\, \boldsymbol{g}^T(-H)^{-1}\boldsymbol{g}\big)$, which is *exactly* linear in $\kappa|\boldsymbol{g}|^2$. Averaging over the isotropic direction of $\boldsymbol{g}$ and over $|\boldsymbol{g}|^2 = v\chi_3^2$ turns $\boldsymbol{g}^T(-H)^{-1}\boldsymbol{g}$ into $v\,\mathrm{tr}(-H)^{-1}$, and $\det(-H)\,\mathrm{tr}(-H)^{-1} = e_2(-H)$, the second elementary symmetric polynomial of the eigenvalues. Hence

$$\mathbb{E}\big[|\det(H + \kappa \boldsymbol{g}\boldsymbol{g}^T)|\,\Theta_+\big] = \sigma_{2,\lambda}^3\Big[\mathbb{E}_x \frac{f(x)}{27} - \tau\,\mathbb{E}_x f_2(x)\Big] + \mathcal{O}(\tau^2), \qquad \tau \equiv \frac{s\,\kappa\,v}{\sigma_{2,\lambda}}, \tag{18}$$

with $x \sim \mathcal{N}(m_\lambda, s_\lambda^2)$ and $f_2(x) \equiv \mathbb{E}[e_2(-\zeta)\Theta_+ \,|\, x]$ for the standardised Hessian ζ. In the BBKS shape variables ($x = \sum \lambda_i$, $y = (\lambda_1 - \lambda_3)/2$, $z = (\lambda_1 - 2\lambda_2 + \lambda_3)/2$, density $\propto y(y^2 - z^2)e^{-\frac{5}{2}(3y^2+z^2)}$, peak condition $x - 3y + z > 0$), $e_2 = x^2/3 - (3y^2 + z^2)/3$, so f_2 follows from the same two-dimensional integral as f; that integral reproduces BBKS's f to 10^{-10}, and $f_2 \to x^2/3 - 1/3$ for $x \gg 1$. The peak density (17) is multiplied by

$$Q = 1 - \tau\, R_2(s\,m_\lambda, s_\lambda), \qquad R_2(m,s) = \frac{27\int_0^\infty f_2\,\mathcal{N}(x;m,s^2)\,\mathrm{d}x}{\int_0^\infty f\,\mathcal{N}(x;m,s^2)\,\mathrm{d}x} \xrightarrow{m\gg 1} \frac{9}{m}. \tag{19}$$

The remainder is a boundary effect. For $\tau > 0$ (a positive semi-definite addition) maxima can only be lost, the exact factor is $\mathbb{E}[\det(-H)(1-\tau\,q)_+]/\mathbb{E}[\det(-H)]$ with $q \geq 0$, and (19) is a lower bound. For $\tau < 0$ the interior part is exactly (19) and configurations that become maxima add a positive $\mathcal{O}(\tau^2)$ piece. Where $\tau R_2 > \frac{1}{2}$ the code continues Q smoothly and positively; such nodes carry no weight in practice.

Size. $\tau \propto v/\sigma_{2,\lambda} \propto \mathcal{A}^{1/2}$ is formally one power of the r.m.s. amplitude below the λ effect, but $\kappa \simeq -2\lambda^2/c$ for the quadratic template, so it is not small when λ is large and negative. $\kappa \equiv 0$ for the logarithmic template. At nodes on the ridge $z = \rho\,\sigma_{0,r}\,c/\sigma_{0,c}$ with $\sigma_{0,c} = 0.1$, $r = 2.74/k_*$ (lognormal, transfer function, cut):

Δ	template	c	MC rank-one effect	$Q-1$, eq. (19)	MC / eq. (17)$\times Q$
1.0	$f_{\rm NL} = -2$	0.60	+0.359	+0.341	1.0085
1.0	$f_{\rm NL} = -2$	0.45	+0.223	+0.219	1.0019
1.0	$f_{\rm NL} = +2$	0.60	+0.0093	+0.0093	0.9997
1.0	curvaton $r_{\rm dec} = 0.3$	0.45	+0.0287	+0.0287	0.9997
1.0	$\mu_* = 5/6$	0.60	0	0	0.9997
0.3	$f_{\rm NL} = -2$	0.60	+0.0401	+0.0401	0.9998
0.3	$f_{\rm NL} = +2$	0.60	+0.0049	+0.0049	0.9999

(Monte Carlo with 10^6 samples, statistical error (3–16) $\times\, 10^{-4}$.) Only the most extreme node, where the base density is already suppressed twentyfold, shows the $\mathcal{O}(\tau^2)$ remainder, at 0.85%; it moves A_* by $\sim 0.02\%$.

8 Result

With $c = C_\ell/F'(z)$, $\lambda = C_\ell F''(z)/F'(z)^2$ and $s = \mathrm{sign}F'(z)$,

$$\beta_{\rm pk}^{\rm 3b} = V\int \mathrm{d}z \int_{C_{\ell,\rm th}}^{2\Phi} \mathrm{d}C_\ell\, \frac{P_{\rm G}(C_\ell/F'(z), z)}{|F'(z)|}\, \frac{G(s\,m_\lambda, s_\lambda)}{(2\pi)^{3/2}R_{*,\lambda}^3}\, \big(1 - \tau R_2\big)\, \mathcal{K}(C - C_{\rm th})^\gamma. \tag{20}$$

In the code Y_λ is normalised by $(1+\lambda^2)^{1/2}$, i.e. $\lambda = \tan\theta$, which leaves $R_{*,\lambda}$, m_λ, s_λ and τ unchanged and keeps $F' \to 0$ finite. Threshold statistics are the same integral without V and without the peak factor. So, within this framework, the two statistics differ only through that factor, and NG enters the difference through $\lambda(C_\ell, z)$, $R_{*,\lambda}$, m_λ and s_λ.

9 Checks

1. $\lambda = 0$ and $F'' = 0$: identical to Phase 3a (unit tests: exact for Gaussian ζ, 7×10^{-7} in $\log\beta$ for $F = 1.3\,\zeta_\mathrm{g}$).

2. Narrow lognormal spectra: $|\log\beta^{3\mathrm{b}} - \log\beta^{3\mathrm{a}}| = 0.0011$ at $\Delta = 0.05$ against 0.032 at $\Delta = 1$ ($f_\mathrm{NL} = 1$, unit test).

3. Conditional Monte Carlo of (4) at fixed (c, z), written independently of Secs. 5–7 (`ngpbh.validation`): sample $\nabla\zeta_r$ from its marginal and weight by the joint density of $(\nabla C_\mathrm{g}, \nabla\zeta_r)$ at $\nabla C_\mathrm{g} = -\lambda\nabla\zeta_r$; sample the twelve Hessian components from their joint Gaussian conditional on (c, z), built directly from (6); evaluate the exact $\nabla\nabla C_\ell$, its determinant and definiteness. Without the rank-one term it reproduces (17) within its statistical errors (2×10^{-4} typically, 2×10^{-3} at the most suppressed node); with it, the table of Sec. 7.

4. Lattice (`scripts/lattice_peaks.py`): Gaussian random fields on a periodic 256^3 grid, lognormal $\Delta = 1$, $r = 8$ cells, $\sigma_{0,c} = 0.15$, 24 realisations; C_g and ζ_r by FFT filtering with the same kernels and cut; local maxima of $C_\ell = F'(\zeta_r)C_\mathrm{g}$ (26 neighbours) counted above $C_\ell = \nu\sigma_{0,c}$. The predictions use the lattice's own mode sums for the moments, so only grid effects separate them from the counts. Lattice/prediction for $\nu = 1.5,\ 2,\ 2.5,\ 3$:

template	Phase 3b, eq. (20)	Phase 3a (maxima of C_g)
Gaussian (grid calibration)	1.010, 0.995, 0.988, 0.976	same
$f_\mathrm{NL} = -2$	1.006, 1.003, 0.976, 0.949	1.183, 1.182, 1.159, 1.139
$f_\mathrm{NL} = +2$	1.011, 1.008, 1.004, 0.997	0.911, 0.936, 0.960, 0.977
$\mu_* = 5/6$	1.007, 1.003, 0.997, 0.998	0.929, 0.942, 0.956, 0.976

   Statistical errors are 0.2–0.6%, and 1–2.7% at $\nu \geq 2.5$ for $f_\mathrm{NL} = -2$. Phase 3b follows the counts to the level of the Gaussian grid trend; the Gaussian-peak approximation misses them by up to 18%. In the required amplitude this is a shift of at most 1.7% (Sec. 10), because $\delta\ln A_* \simeq \delta\ln n/|\ln\beta|$.

10 Consequence for the abundance

Same threshold (Musco et al. with the transfer function), window, transfer function, cut, $\mathcal{K} = 3.3$, $\gamma = 0.36$ and $k_* = 10^7\,\mathrm{Mpc}^{-1}$ for all statistics (`results/phase2_amplitudes.csv`):

- $A_*^{3\mathrm{b}}/A_*^{3\mathrm{a}} = 0.983$–$1.000$ for $\Delta = 0.1$–2, $|f_\mathrm{NL}| \leq 2$, $\mu_* \in \{5/6, 5/2, 10, -5/2\}$: largest for $f_\mathrm{NL} = -2$ at $\Delta = 0.5$.

- $R = A_*^\mathrm{peaks}/A_*^\mathrm{threshold} = 0.770$–$0.789$ for Gaussian ζ and 0.738–0.817 for $|f_\mathrm{NL}| \leq 2$; the double ratio R/R_Gauss lies in 0.958–1.036, and reaches 1.16 for $\mu_* = 5/6$ at $\Delta = 2$.

11 Not included (common to both statistics)

The spherical-symmetry step behind (3), which Young [2] questions for broad spectra (high peaks of C need not be high peaks of ζ); the condition that r maximises the compaction (prescription (iii)); profile dispersion around the mean profile (Escrivà 2607.08738); type-II perturbations. All of them affect threshold statistics in the same way, so the comparison stays like for like.

References

[1] G. Ferrante, G. Franciolini, A. Iovino, A. Urbano, arXiv:2211.01728.

[2] S. Young, arXiv:2201.13345.

[3] J. M. Bardeen, J. R. Bond, N. Kaiser, A. S. Szalay, Astrophys. J. 304 (1986) 15.

[4] C. Germani, M. A. Gorji, M. Uwabo-Niibo, M. Yamaguchi, arXiv:2503.05434 (peaks of a local function of *one* Gaussian field, where peak positions coincide).